

Once LokiStack is deployed, you should observe the pods running successfully, as illustrated in Figure 11.

Deploying the logging instance with LokiStack as the log store

To utilise LokiStack as the log store, you can create a cluster logging instance and specify LokiStack as the desired log storage option. Figure 12 depicts an example YAML configuration for a cluster logging instance.

In the configuration, ensure to specify LokiStack as the log store type, along with the name of your LokiStack instance. Apply the configuration using the following command:

```
oc apply -f <cluster-logging-instance.yaml>
```

Upon successful application of the configuration, you can verify the deployment by checking that the collector pods are running as expected in Figure 13.

Deploying and monitoring application logs with the logging operator

The application we deployed is a simple BusyBox container, generating a series of counter logs. Once the application is deployed, navigate to the 'Operators' section and select the logging operator. There, you'll find an option to enable the console plugin. Make sure to enable it. Now, head over to the 'Observe' section on the console. Click on the

```
[root@bastion lokistack-automation]# oc get po
NAME                                READY   STATUS    RESTARTS   AGE
cluster-logging-operator-599c9495cc-479t2    1/1     Running   0           44h
lokistack-sample-compactor-0                1/1     Running   0           2m24s
lokistack-sample-distributor-68878fbbd5-kh5zf  1/1     Running   0           2m27s
lokistack-sample-distributor-68878fbbd5-vnclh  1/1     Running   0           2m27s
lokistack-sample-gateway-59dff8bcd5-pr6nk     2/2     Running   0           2m21s
lokistack-sample-gateway-59dff8bcd5-r7plm     2/2     Running   0           2m21s
lokistack-sample-index-gateway-0              1/1     Running   0           2m23s
lokistack-sample-index-gateway-1              1/1     Running   0           106s
lokistack-sample-ingester-0                   1/1     Running   0           2m26s
lokistack-sample-ingester-1                   1/1     Running   0           80s
lokistack-sample-querier-6746f4b74d-7l1d2     1/1     Running   0           2m25s
lokistack-sample-querier-6746f4b74d-8qtwz     1/1     Running   0           2m26s
lokistack-sample-query-frontend-6b75d8b954-6kbc2 1/1     Running   0           2m24s
```

Figure 11: Successful deployment of LokiStack pods

```
apiVersion: "logging.openshift.io/v1"
kind: "ClusterLogging"
metadata:
  name: "instance"
  namespace: openshift-logging
spec:
  managementState: "Managed"
  logStore:
    type: "lokistack"
    lokistack:
      name: lokistack-sample
  collection:
    type: "vector"
```

Figure 12: Example YAML configuration for a cluster logging instance

'Logs' option, and you'll be able to view the logs as described above. In Figure 14, we can see the application logs in the dashboard.

Use cases

Centralised logging

Scenario: An organisation manages a distributed system comprising numerous microservices deployed on s390x architecture. To streamline troubleshooting and monitoring efforts, it opts to centralise log data from disparate sources.

Solution: By deploying Loki on Kubernetes using the Loki Operator, the organisation establishes a centralised log aggregation platform. This enables it to efficiently collect, store, and analyse logs from diverse microservices, facilitating

```
[root@bastion ~]# oc get po -n openshift-logging
NAME                                READY   STATUS    RESTARTS   AGE
cluster-logging-operator-6ff4d7b8c6-zzxd8    1/1     Running   0           34h
collector-5n9zz                            1/1     Running   1           15d
collector-6xztr                             1/1     Running   0           15d
collector-dmtbw                             1/1     Running   0           15d
collector-fb7mf                             1/1     Running   1           15d
collector-vm4w4                             1/1     Running   1           15d
collector-wc6f4                             1/1     Running   0           15d
logging-view-plugin-85fc8ffbff-cz676        1/1     Running   0           24h
lokistack-sample-compactor-0                1/1     Running   0           27h
```

Figure 13: Successful deployment of the cluster logging instance with LokiStack